Mohammedzaki I Kochargi

Software Developer

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Github - https://github.com/mohammedzakikochargi

Professional Summary

Highly skilled software engineer with **2** years of experience in developing **Open Source computer vision** pipelines in **C++, C , CUDA** and CMake on **Linux**, **Windows**, and **ARM** platforms for multithreaded **image** and **video framework**. Proficient in leveraging **OpenCV, FFmpeg, Gstreamer** with **GPU** acceleration to deliver efficient and high-performance solutions. Have Handson experience on working **Nvidia tenssorrt** framework Collaborative and innovative, exceeding expectations in cross-functional teams. Continuously focused on mastering emerging technologies for project success.

Skills

C++, C , CUDA, Python OpenCV, FFMPEG, Gstreamer,GPU Image and Video Processing OOPs, Multithreading and OOP's, GDB Debugger, Gtest, design. Design Patterns, Problem solving Motion estimation Cmake, RTSP, Docker

Windows, Linux and ARM . Development on Nvidia Jetson devices STL library, Data structures and BOOST lib Framework and pipeline level development Unit testing H.264 Video Codec, Mp4 Video format
Nvidia tensorrt, Tensorflow, pytorch
Vcpkg, Redis DB

Experience

Apra Labs, Bangalore

2022 - 2023

Software Engineer

03-01-2022 - Current

Tensorrt - Object detection

I have extensive experience with TensorRT, including converting the YOLOv5 model to be compatible with TensorRT for efficient object detection. In this process, I developed CUDA and C++ code to create a custom TensorRT engine, tailoring the inference pipeline for optimal performance. This involved leveraging TensorRT's capabilities to optimize model execution, enhance processing speed, and improve overall detection accuracy by making full use of GPU resources. My work ensured that the object detection system could handle high-throughput scenarios effectively.

Cuda Kernel:

I have experience in writing CUDA kernels for high-performance applications. This expertise allows me to optimize computational tasks and leverage GPU acceleration to achieve significant performance improvements in data processing and other intensive operations.

virtual camera stack: -

I worked on a project where I was responsible for building the backend video pipeline using ApraPipes, **GStreamer**, and **Redis** DB. I gained expertise in GStreamer **RTSP** streaming, **appsrc**, **H264Enc**, **VideoConvert**, client/server connections, Linux camera device trees (/dev/video), and **appsrc**. The virtual cameras, which are ONVIF compliant, can be discovered by **ONVIF Device Manager** (ODM). This product eliminated the need for hundreds of physical cameras for development and was tested on **Windows**, **Linux**, and **macOS**. The entire product was deployed using Docker, **Docker containers, images, Docker Compose**, and Dockerfiles.

Network Video Recorder (NVR) :

I was involved in the development of a Network Video Recorder (NVR) product, where I gained valuable experience in designing and creating the user interface (UI) using the **GTK** framework. I developed the complete heavy backend pipeline, which involved NVIDIA, **OpenCV**, and **FFmpeg**. This backend pipeline included transformations, **color changes**, **memory conversions** for **GPU** computation, **H.264 encoding and decoding**, as well as MP4 reading and writing for playback. I also developed modules for **RTSP** pulling and RTSP streaming

H.264 Dynamic Forward and Reverse Playback:

I designed a dynamic **reverse** playback feature for **H.264** encoded videos/bitstreams that required intricate modifications to Nvidia's decoder. This enhancement allowed for the decoding of a Group of Pictures (GOP) and the subsequent reversal of the decoded frames. The solution was adept at managing scenarios involving abrupt playback direction changes and seamless seeking.

Junior software developer

Proficiently developed computer vision pipelines using C++, CMake, and CUDA for accelerated processing, ensuring high-speed performance.

Motion Estimation detection:

Successfully designed and implemented a **motion vector extraction** pipeline while **h264 encoding** and **decoding(Without frame reconstruction)**, enhancing video processing capabilities.

MP4 Container:

I have a strong understanding of the MP4 container, particularly in reading and writing, which I've applied in video playback pipelines across various products. This expertise has been crucial in ensuring smooth video playback and efficient handling of multimedia content.

Generic Overlay Module:

Developed a versatile image **overlay** module, allowing seamless integration of graphical elements into videos and images using **OpenCv**.

Multimedia Queue Functionality:

Spearheaded the creation of a multimedia queue to cache images, optimizing data processing and enhancing pipeline efficiency.

Generic Color Conversion Module:

mplemented a generic color conversion module using OpenCV, enhancing image processing capabilities and enabling seamless color transformations of image types such **RGB**, **YUV**, **Bayer** and **MONO**

H.264 Decoder:

Have developed cross platform H.264 decoder module which runs on Win, Lin and arm64 which **Nvidia's video codec sdk** and **Jetson multimedia Api**

Face Detection:

I have experience with face detection techniques, including using **Haar cascades and SSD (Single Shot Detector)**. These methods have allowed me to efficiently detect faces.

Software Trainee At Apra Labs

C++

Proficient in Framework-level development with a focus on interconnected pipelines using modules. Developed image processing modules during training.

Education

Bachelor of Engineering in Computer science - Gogte Institue of Technology *2018 - 2022*

CGPA - 8.05

Board Member - Rotaract club of GIT Sports

Incharge - Association of Computer Engineers.

Achivements

Developed a motion estimation module which extracts motion vectors from a H.264 bitstream while decoding without frame reconstruction, Hence resulting in fastest way of motion vector extraction with minimum CPU usage. Ultimate team player award at Apra Labs . Contributed a lot on open source

Languages

English
Hindi
Kannada
Marathi